Project title: Sound intensity with microphone array

Authors: Sokrates Kjølner, Patryk Ryczek and Kasper Ng Bergan

Abbreviations

dBFS: Decibels relative to Full Scale

MCU: Microcontroller Unit

TDM: Time Division Multiplexed

SNR: Signal to Noise Ratio

RAM: Random-access Memory

SRAM: Static random-access Memory

DTCM: Data Tightly-Coupled Memory

SAI: Serial Audio Interface

VDD: Positive supply voltage

GND: Ground

WSO: Word Select Output

WS: Word Select

SD: Serial Data

SCK: Serial Clock

DMA: Direct Memory Access

UART: Universal Asynchronous Receiver-Transmitter

CPU: Central Processing Unit

BLE: Bluetooth Low Energy

LoRaWAN: Long Range Wide Area Network

SPL: Sound Pressure Level

FS: Full Scale

GUI: Graphical User Interface

Background

This project demonstrates a small microphone array that measures sound intensity in dBFS, sends the result wirelessly and plots it in real time. Two microphones are used to keep the project simple and within the time available. The NUCLEO-L476RG is also limited: it has only 128 KB of SRAM and no cache or DTCM, so complex processing of many microphones would quickly run out of memory.

The STM32L476RG reads data from two ICS-52000 microphones through the SAI peripheral with DMA. A double (ping-pong) buffer prevents data from being overwritten while the CPU is processing it: the CPU works on one half while DMA fills the other, and they swap when both are done. The CPU converts each block of raw samples into a dBFS value, which is sent over UART to an nRF52840 module and forwarded over Bluetooth to a computer for plotting.

The expected outcome is a plot with two graphs that react to the microphones in real time. In a quiet room the level should be around -60 dBFS and rise when sound is picked up. Because the microphones are placed close together, the graphs should look nearly identical, but tapping one microphone should make only its graph spike.

Materials and methods

The system uses two ICS-52000 MEMS microphones, a NUCLEO-L476RG and an nRF52840 BLE module.

The ICS-52000 is a bottom-port digital microphone with a built-in analog-to-digital converter, 65 dBA SNR and a frequency range of 50 Hz to 20 kHz. Its TDM interface carries up to 16 microphones on one data line, which removes the problem of unsynchronized clocks between channels. [1] Together with its low price, this makes it well suited to arrays. Table 1 lists its pins.

Pin Name	Type	Function
VDD	Power	Power, 1.62V to 3.63V. This pin should be decoupled to GND with a 0.1 µF capacitor.
GND	Power	Ground. Connect to ground on the PCB
CONFIG	Input	Pull to VDD. The state of this pin is used at power-up.
WSO	Output	WS output, connect to WS of the next ICS-52000 in the daisy-chain.

WS	Input	Serial Data-Word Select for TDM Interface
SD	Output	Serial Data Output for TDM Interface. This pin tri-states when not actively driving the appropriate output channel. The SD trace should have a 100 kΩ pulldown resistor to discharge the line during the time that all microphones on the bus have tri-stated their outputs
SCK	Input	Serial Data Clock for TDM Interface

Table 1

The NUCLEO-L476RG has an ARM Cortex-M4 MCU with 1 MB flash and 128 KB SRAM. The peripherals used are:

- **SAI:** receives microphone data, since it supports TDM.
- **DMA:** moves SAI and UART data without CPU involvement.
- **Timer:** controls the timed startup sequence the ICS-52000 requires.
- **UART:** sends results to the nRF52840. UART was chosen over SPI and I2C because only a small amount of data is sent after the dBFS calculation, so 115200 baud is sufficient.

The nRF52840 is a multiprotocol module with a 64 MHz Cortex-M4F, 1 MB flash, 256 KB RAM and interfaces including UART, SPI, I2C and I2S. [2] BLE was chosen over LoRaWAN because it offers much higher data rates (up to 2 Mbps versus under 50 kbps), and the application does not need long-range data transfers. [3]

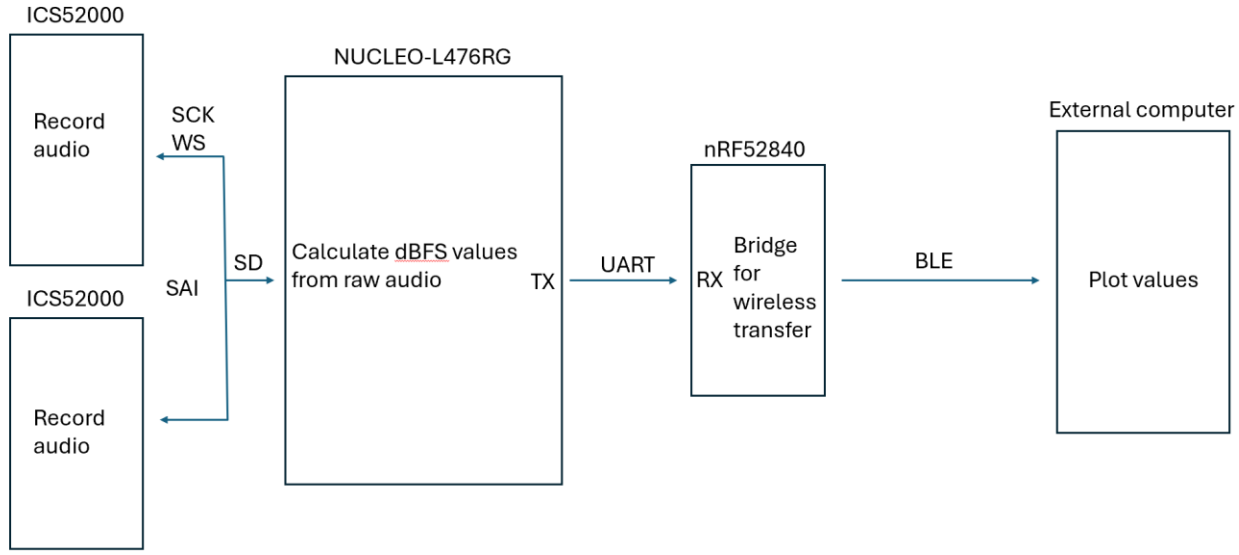
The STM32 firmware is written in C using the HAL library. Main c files are ics52000.c and ics52000.h, these are the driver files which handles everything to do with the ICS52000. The protocols coded in STM32 are SAI and UART. SAI sends microphone data to MCU, this protocol was chosen because it supports TDM. UART is used to send dBFS values from the MCU to the BLE module, this was chosen because there is no need for high data-rate.

For programming the nRF52840 we will use Arduino IDE. This is because of the ease of use. Since nRF52840 does no calculations, it just sends data further there is also no need for optimized code for this MCU. The protocols used with the nRF52840 are UART and BLE. UART is used for receiving data

from the NUCLEO-L476RG and the BLE is used to send data wirelessly to an external computer.

The output of the microphones is shown using a GUI made in python. We use Python for this case because of its broad range of libraries, documentation and the large amount of open-source written code.

System diagram



System requirements/specifications

The system measures sound level in dBFS with digital MEMS microphones, calculates it on the MCU and sends it wirelessly to a computer for plotting.

Power

The NUCLEO is powered over USB. Its 3V3 pin powers the microphones, and the nRF52840 can use 3V3 or 5V.

Parameter	**Conditions**	**Min**	**Typ**	**Max**	**Units**
ICS52000					
Supply voltage		1.62		3.63	V
Supply Current (VDD = 3.3V)	Normal		1.1	1.5	mA

nRF52840					
Supply voltage		1.7		5.5	V
Supply current	CPU active, 1Mbps Bluetooth		8.1		mA
NUCLEO-L476RG					
Supply Voltage	Standard operating voltage	1.71		3.6	V
Supply current	Range 1 25 C		10.2	11.22	mA

Table 2 Supply requirements

Microphone

The microphones must deliver a continuous, high-precision stream to the MCU.

Parameter	Conditions	Min	Typ	Max	Units
Sample rate		7.19		52.8	kHz
Directionality			Omni		
Sensitivity	1 kHz, 94 dB SPL	-27	-26	-25	dB FS
SNR			65		dBA
Noise Floor	20 Hz to 20kHz, A-weighted, rms		-91		dB FS

Table 3 ICS-52000 performance [4]

Hardware needed for the ICS-52000 circuit is one 0.1 uF decoupling capacitor on VDD and a 100k ohm pull down resistor on the SD line to discharge the line during the time that all microphones on the bus have tri-stated their output.

Data rate

When calculating what data-rate needed it is important to note that the values given in the datasheet are theoretical values. Even though the datasheet suggests that the nRF52840 can send data 2Mbps, this is only under perfect circumstances, a lot less should be expected.

The sample rate is 16 kHz. The ping-pong buffer holds 512 words, and each half is processed as one block. With two microphones words per second is 32 000 (Sample rate * number of mics), dBFS values per second is 250 ((words per second / block) * number of mics). Each dBFS value transferred over UART is 40 bits in size (1 start bit + 8 data bit + 1 stop bit) * number of bytes. The theoretical absolute minimum data transfer speed is 10 kbps (dbfs values per second * size of uart transfer). 10 kbps is way lower than the ratings for UART and BLE, therefore data transfer should not be a bottleneck.

Parameter	**Conditions**	**Min**	**Typ**	**Max**	**Units**
Frequency deviation	2Mbps transfer		+/- 500		kHz
UART Baud rate				1000	kbps

Table 4 Data transfer speed [5]

Processing

The MCU must receive samples via SAI and calculate dBFS continuously while transmitting results over UART. Both transfers use DMA to keep the CPU free.

Parameter	**Conditions**	**Min**	**Typ**	**Max**	**Units**
SAI Frame length				256	bit
UART Baud rate				10	Mbit/s
Clock speed		100kHz		80	MHz

Table 5 NUCLEO-L476RG specifications [6]

References

[1] Cirrus Logic, “TDM Audio Interface: A Tutorial,” *EDN*, Sep. 1, 2014. [Online]. Available: https://www.edn.com/tdm-audio-interface-a-tutorial/

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[3] Gaotek, “Comparison of BLE and LoRaWAN,” Gaotek. [Online]. Available: https://gaotek.com/comparison-of-ble-and-lorawan/

[4] TDK InvenSense, “ICS-52000 Datasheet,” Doc. DS-000121, Rev. 1.3, TDK Corporation. [Online]. Available: https://product.tdk.com/system/files/dam/doc/product/sw_piezo/mic/mems-mic/data_sheet/ds-000121-ics-52000-v1.3.pdf

[5] Nordic Semiconductor, “nRF52840 Product Specification: Key Features,” Nordic Docs. [Online]. Available: https://docs.nordicsemi.com/r/bundle/ps_nrf52840/page/keyfeatures_html5.html

[6] STMicroelectronics, “NUCLEO-L476RG: STM32 Nucleo-64 development board with STM32L476RG MCU,” STMicroelectronics. [Online]. Available: https://www.st.com/en/evaluation-tools/nucleo-l476rg.html